

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 05.1

Design and Analysis of an FIR High-Pass Filter Using the Rectangular Window Technique in MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000;          % High-pass cutoff frequency (Hz)
fs = 8000;          % Sampling frequency (Hz)
N = 50;             % Filter order
wc = fp/(fs/2);     % Normalized cutoff (0 to 1)
%% --- FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);
%% --- Frequency Response Plot ---
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

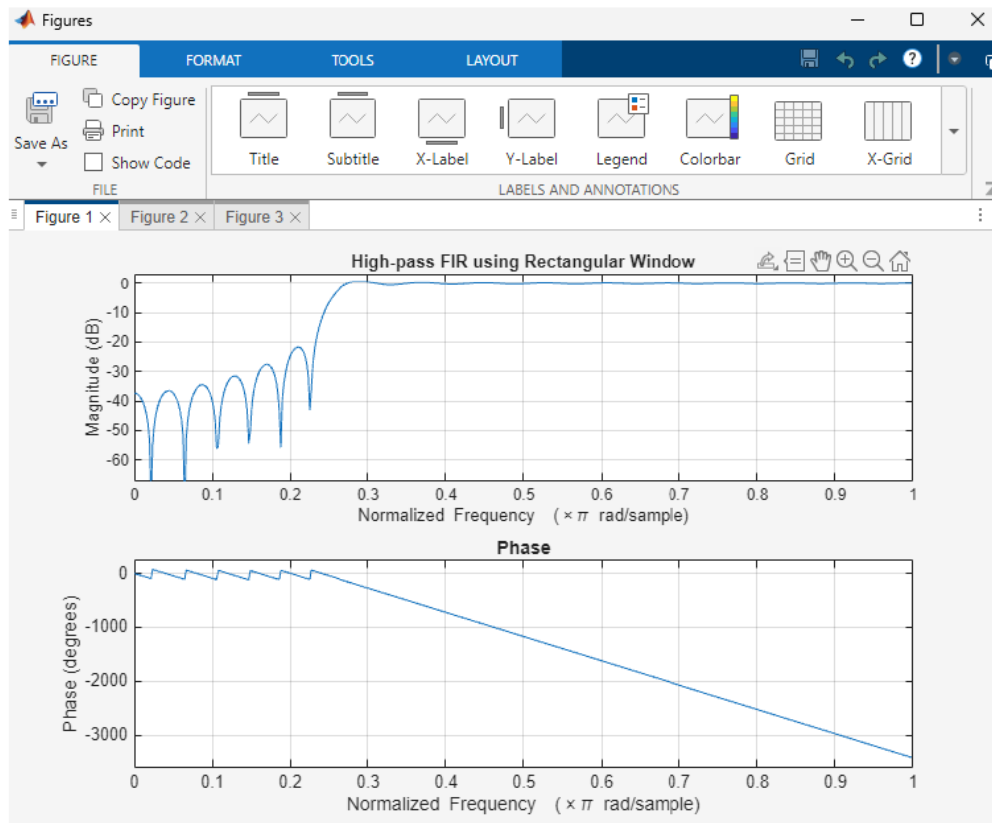
%% --- Frequency Response Plots ---
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

OUTPUT:

```
--- High-Pass Coefficients (Rectangular Window) ---
-0.0089
```

-0.0000
0.0097
0.0143
0.0106
-0.0000
-0.0117
-0.0175
-0.0131
-0.0000
0.0148
0.0225
0.0171
-0.0000
-0.0202
-0.0314
-0.0247
-0.0000
0.0318
0.0524
0.0445
-0.0000
-0.0741
-0.1572
-0.2223
0.7407
-0.2223
-0.1572
-0.0741
-0.0000
0.0445
0.0524
0.0318
-0.0000
-0.0247
-0.0314
-0.0202
-0.0000
0.0171
0.0225
0.0148
-0.0000
-0.0131
-0.0175
-0.0117

-0.0000
0.0106
0.0143
0.0097
-0.0000
-0.0089



Result

1. The FIR High-Pass Filter was successfully designed using the Rectangular Window technique for the specified cutoff frequency, sampling frequency, and filter order.
2. The FIR filter coefficients were successfully generated using MATLAB and the designed filter was implemented.
3. The magnitude response and phase response of the designed FIR High-Pass Filter were obtained and analyzed. The magnitude response showed attenuation of low-frequency components and passage of high-frequency components, confirming the desired high-pass

filtering behavior. The phase response exhibited the expected linear-phase characteristics of an FIR filter.

Practical Question 1

Design an FIR High-Pass Filter using the Rectangular Window technique in MATLAB for a sampling frequency of 10000 Hz, cutoff frequency of 2500 Hz, and filter order of 50.

Tasks:

1. Calculate the normalized cutoff frequency.
2. Generate the FIR filter coefficients using the `fir1()` function with a Rectangular Window.
3. Plot the magnitude response and phase response of the designed filter.
4. Verify that low-frequency components are attenuated and high-frequency components are passed.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

Design and implement an FIR High-Pass Filter using the Rectangular Window technique in MATLAB for a sampling frequency of 16000 Hz, cutoff frequency of 4000 Hz, and filter order of 60.

Tasks:

1. Design the FIR High-Pass Filter using the specified parameters.
2. Generate and display the FIR filter coefficients.
3. Obtain the magnitude response and phase response using the `freqz()` function.
4. Determine the cutoff frequency and examine the transition region.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To design an FIR high-pass filter using a rectangular window and analyze its magnitude and phase response.

Student Details

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